Course Name: Computer Networks

Course Outcome

CO1: Understand the concept of OSI and TCP/IP reference model
CO2: Understand the basics of data transmission at physical layer.
CO3: Understand the channel allocation using ALOHA, CSMA and CSMA/CD.
CO4: Apply error detection and correction technique to eliminate transmission error.
CO5: Analyze the fixed and variable length address (IPv4) subnetting for the given scenarios.
CO6: Understand the design issues of the transport layer.
CO7: Understand the mechanism of protocols at application layer such as FTP, HTTP, Telnet, DNS.
CO8: Understand IPv6 addressing and differentiate it from IPv4.

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Mid Term Examination, EVEN Semester 2023-24

B.Tech. (CSE/AIML/CCV/DA/CSF/IOT), 2nd Year, IV Semester

BCSC0008 Computer Networks

Time: 2 Hours **Maximum Marks: 30**

Section – A

Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1.	For each of the following four networks, discuss the consequences if a connection fails. a. Five devices arranged in a mesh topology b. Five devices arranged in a star topology (not counting the hub) c. Five devices arranged in a bus topology d. Five devices arranged in a ring topology	3	2	U	F
2	A pure ALOHA network transmits 200-bit frames on a shared channel of 200 kbps. What is the throughput if the system (all stations together) produces? a. 1000 frames per second b. 500 frames per second c. 250 frames per second	3	3	A	P
3	Compare and contrast a circuit-switched network and a packet-switched network.	3	2	A	C
4	Tom is the network administrator for a medium-sized company. The company is planning to upgrade its network infrastructure, and Tom needs to decide on the transmission media to use for connecting various	3	1	An	C

	devices. He is considering options like twisted pair, fiber optics, and co-axial cable. Identify the pros and cons of above mentioned transmission media?				
5	Identify different transmission impairments that affect data communications in detail. Consider an extremely noisy channel in which the value of the signal-to-noise ratio is almost zero. In other words, the noise is so strong that the signal is faint. For this calculate the channel capacity.	3	2	A	C

Section – B

*Attempt **All** Questions*

5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Compare and contrast the Go-Back-N ARQ Protocol with Selective-Repeat ARQ. In a Go-Back-N ARQ (Automatic Repeat Request) protocol, a sender transmits frames to a receiver over a noisy channel. The sender can transmit up to 7 frames before waiting for an acknowledgment (ACK) from the receiver. If a frame is lost or corrupted, the receiver discards it, prompting the sender to retransmit all frames starting from the one that was lost. Assume the sender has already sent frames 1, 2, 3, and 4 successfully, and frame 5 gets corrupted during transmission. If the sender is using a Go-Back-N ARQ protocol, how many frames does the sender retransmit in this scenario? Using 5-bit sequence numbers, what is the maximum size of the send and receive windows for each of the following protocols? a. Go-Back-NARQ b. Selective-Repeat ARQ	5	3	A	P
7	A small start-up company is experiencing network connectivity issues within their office based on OSI layers. They suspect that the problem lies within the OSI model layers. As a network technician, how would	5	2	An	C

	you approach troubleshooting this issue considering different services of OSI layer?				
8	A. How the simple parity check is related to the two-dimensional parity check? B. Assume that sender sent the encoded data using even parity hamming code (single bit detection and correction). The receiver received the bit stream as 001011110101. Answer the following: (Assume right most bit as 1st bit in the stream) i. How many check bits (redundant bits) are there in the received bit stream? ii. Is there any error in the received data? If yes, write the bit position which is in error? Show your calculation. Write the data stream (without check bits) generated by sender	1+4	4	A	M